

Sonali Srivastava

Ghaziabad, Uttar Pradesh | sonali200530@gmail.com | +91 8058833750 | Portfolio | LinkedIn | GitHub

Summary

Electronics and Instrumentation undergraduate skilled in IoT and embedded systems, sensor interfacing, Arduino programming, serial communication, and real-time data acquisition.

Education

Banasthali Vidyapith , BTech in Electronics and Instrumentation	July 2023 – Present
• CGPA: 7.73/10.0	
Ascent International School , 12th Grade	July 2022– July 2023
• Percentage: 65.5%	
Ascent International School , 10th Grade	July 2020– July 2021
• Percentage: 87%	

Projects

CoalShield	CoalShield
• Developed CoalShield, a real-time smart safety jacket for coal miners designed to enhance worker safety through continuous environmental and physiological monitoring for faster hazard detection and emergency response.	
• Languages: Embedded C/C++	
• Tech Stack: putty, VS Code	
• Peripherals: Raspberry Pi 4 B+, Motion/Posture Sensor, Temperature & Humidity Sensor, Heart Rate Sensor, Oxygen Monitoring Sensor, Buzzer	
SENTRY-X	Sentry-X
• Developed a real-time intrusion detection system to address delays in manual surveillance by integrating Arduino-based sensor interfacing with Python-based verification logic.	
• Languages: Embedded C++, Python	
• Platform: Arduino	
• Protocols & I/O: I2C, Analog & Digital I/O (GPIO)	
• Peripherals: Ultrasonic Sensor, Potentiometer, Buzzer	
• Tools: Arduino IDE, Visual Studio Code, Serial Monitor, Git	

Skills

Programming: C++, Python, PLC Programming, Arduino Programming

Hardware: Arduino, Raspberry Pi, UART, I2C, Serial Communication, Microcontroller interfacing

Tools/Software: Arduino IDE, MATLAB, Putty, Simulink, Figma, GitHub, TIA Portal, RoboAnalyzer, Kuka Sim

Leadership Activities

Core Member Team Samarthya(Electronics Club)	November 2023-Present
• Organized ENGINEER'S DAY for project demonstration	
• Organized 'Circuit-o-Geek', a circuit-making competition.	
• Led 'Squid Game' at Mayukh Fest, an electronics-themed fun event.	
• Assisted in 'Pathway to Professional World' with industry experts.	
• Contributed to 'Photon 3.0', Team Samarthya's technical magazine	